

Project Summary

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IT 106 - DL1

GMU Campus Budget & Calorie Meal Planner

Overview

This is a program that helps GMU students plan daily meals within a budget. Users enter a daily budget and gender, which sets their calorie goal (2,000 for female, 2,500 for male). The program lets users pick breakfast, lunch, or dinner repeatedly, tracking spending and calories. The day ends when the user stops, runs out of budget, or reaches their calorie goal. If they go over, the program calculates jogging miles needed to burn off extra calories (100 cal per mile).

Design

Five separate classes:

- **MealData**: Stores all meal info as parallel arrays (names, costs, calories)
- **BudgetTracker**: Manages remaining budget and affordability checks
- **CalorieTracker**: Tracks daily calories and checks against the goal
- **UserInterface**: Handles all input prompts and output messages
- **MealPlanner**: Main driver; controls program flow and calls the other classes

Each class has one job, keeping code simple and easy to follow.

Key Implementation Details

- Parallel arrays for meal data (each index represents one complete meal)
- Input validation in loops (reject invalid budget, gender, meal type, choice)
- Cheapest meal (\$5) acts as the affordability threshold
- Loop stops automatically when budget is too low or goal is reached
- Final summary shows total calories vs. goal, with jogging miles if over

Testing

Tested edge cases: exact budget match, hitting goal exactly, going over goal, running out of budget, unaffordable meal, and invalid input at each prompt. All scenarios work as specified.